NISHAN P

Java Backend Engineer

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LinkedIn — Github — Portfolio

Profile

Software Engineer with nearly 3 years of experience in Java backend development, specializing in scalability, performance optimization, Agile Scrum, and GenAI integration (RAG, LLM APIs). I have developed, maintained, and documented critical applications in the **stock market** and **financial security** domains. Proven in collaborating with cross-functional teams to deliver scalable, high-performance solutions aligned with business goals. Passionate about emerging technologies, I continuously solve complex software challenges.

Skills

- **Languages & Databases:** Java, MSSQL, MYSQL, MongoDB
- **Frameworks Web:** Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Security, Spring AOP, WebSocket, Java Swing, HTML, CSS, Thymeleaf
- **API & Security:** REST APIs, SOAP APIs, GraphQL, JWT Authentication, Swagger
- **Caching & Messaging:** Kafka, Redis, Caffeine
- **Core Concepts:** Microservices Architecture, Distributed Systems, Multithreading, Concurrency, Design Patterns, System Design, Performance Optimization, Event-Driven Architecture
- **Tools:** Docker, Git, Maven, Gradle, Azure DevOps, Postman, Openshift
- **Observability & Monitoring:** Elasticsearch, Kibana, Log Aggregation
- **AI & ML:** OpenAI, Gemini API, Grok API, Claude (Anthropic), Ollama (Local LLM Deployment), RAG, Prompt Engineering; Completed Claude 101, Building with Claude API, Anthropic CTF
- **Testing:** JUnit, TestNG, Selenium, WireMock, Unit and Integration Testing

Experience

Geojit Technologies Pvt. Ltd. — Kochi, India
Software Engineer — Nov 2023 – Present

- Migrated 7–8 legacy Core Java (Ant-based) applications into 21 modern Spring Boot + Maven projects, improving maintainability, scalability, and CI/CD adoption
- Optimized backend codebase, reducing execution time by **20%** and enhancing overall system performance
- Resolved **98%** of reported issues across backend and frontend, performing continuous production troubleshooting to reduce downtime, improve system stability, and accelerate release cycles
- Built and optimized real-time data pipelines using **Apache Kafka**, supporting high-throughput financial transactions
- Implemented **Redis** caching for high-frequency database operations, improving throughput by **15%** and reducing database load
- Applied DAO and MVC design patterns with DTOs, ensuring code quality and security using SonarQube and Trivy, and leveraging AI-assisted development tools (GitHub Copilot, ChatGPT, Claude) to improve productivity
- Implemented Docker-based containerization, enabling consistent deployment across development and production environments
- Participated in Agile Scrum processes including sprint planning, daily stand-ups, and peer code reviews to ensure timely delivery

Projects

Payout Processing System

- Designed and developed a distributed payout processing system using **Kafka**-driven event orchestration and microservices architecture to handle high-volume financial transactions
- Implemented end-to-end workflow including validation, BP blocking, voucher creation, duplicate checks, bank balance validation, and bank API integration with failure handling
- Ensured reliable and scalable processing with asynchronous pipelines, handling concurrent requests and maintaining transactional consistency across systems

Esigner (ProteanTech)

- Built secure eSign workflow with HTML/PDF → Base64 document processing and external API integration, implementing **AES encryption**, **hashing**, and **AOP**-based request decryption for data integrity and validation
- Designed asynchronous callback handling with caching and DB persistence, ensuring reliable document status tracking under concurrent load
- Resolved critical Java heap space issues caused by in-memory Base64 storage by implementing bounded caching and cleanup strategies, stabilizing system performance

BackOffice Assist (Job Scheduling System)

- Built centralized scheduling platform automating FTP/SFTP, DB procedures, and API workflows with UI-based job management and retries
- Implemented asynchronous execution using **CompletableFuture** with **RetryTemplate**-based retry logic and failure notifications
- Integrated **RBAC** with **JWT-based authentication**, enabling workflow-level access control across the system
- Handles **50+ workflows daily** with **99% success rate**, reducing manual effort

OTP Messaging Service

- Built scalable system processing **100K+ daily SMS & email messages** with multi-provider (TATA SMPP, ACL REST, Karix REST) support
- Implemented failover and instant resend mechanisms for high availability
- Deployed on **AWS infrastructure** with multiple instances, enabling dynamic provider switching using **Redis Pub/Sub** for zero-downtime updates

MCX Exchange Connector

- Worked on a TCP-based exchange integration using Apache MINA and TIBCO that processes order requests from OMS and communicates with exchange systems
- Implemented parallel processing and auto-reconnect mechanisms, eliminating **10–15 minute downtime** and improving system reliability

CDSL Account Opening

- Replaced scheduler-based workflow with API-driven sequential processing using **Resilience4j** (Retry, Circuit Breaker, Rate Limiting)
- Reduced processing time from **15 min** → **3 min**, improving system responsiveness and reliability

CRD Automation

- Developed Java-based Selenium/TestNG automation replacing manual QA workflows, reducing execution time from **10 min** → **2 min** with reporting

Education

Master of Computer Applications (MCA) 2021 – 2023
APJ Abdul Kalam Technological University, Kerala